

Multiple levels of replica symmetry breaking

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Parisi measures

Given a (random) function

$$\mathcal{H} : \Sigma \rightarrow \mathbb{R}$$

and $E \in \mathbb{R}$, what can we say about the level sets

$$\mathcal{L}(E) := \{\sigma \in \Sigma : \mathcal{H}(\sigma) = E\}?$$

A classic problem

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A classic problem

$$\mathcal{H} : \Sigma \rightarrow \mathbb{R}, \quad \mathcal{L}(E) := \{\sigma \in \Sigma : \mathcal{H}(\sigma) = E \pm \varepsilon\}.$$

Some ways to address the question:

- $\Sigma = \Sigma_N$ a metric space, $\mathcal{H} = \mathcal{H}_N$; $\varepsilon = \varepsilon_N$ small.
- Choose two points in the level set $\mathcal{L}(E)$ and say something about their distance.
- Topology of the level set. Is $\mathcal{L}(E)$ connected? How many connected components? How many holes?

A picture

energylandscape.jpg

A thermodynamical approach

Fix $\beta > 0$, the Gibbs measure associated to $\mathcal{H}_N$ is defined as

$$G_{N,\beta}(d\sigma) = \frac{\exp(\beta\mathcal{H}_N(\sigma))d\sigma}{Z_N(\beta)},$$

where

$$Z_N(\beta) := \int_{\Sigma} \exp(\beta\mathcal{H}_N(\sigma))d\sigma.$$

Denote by $\langle \cdot \rangle_{\beta}$ the Gibbs average with respect to the Gibbs measure $G_{N,\beta}$

$$F_N(\beta) = \frac{1}{N} \log Z_N(\beta).$$

Equivalence of ensembles

There exists a map between temperature parameter and energy levels:

$$\beta \mapsto E$$

Let $\beta > 0$ be fixed. Assume for some $\delta > 0$ there exists a nonrandom function $F : (\beta - \delta, \beta + \delta) \rightarrow \mathbb{R}$ such that for any $\beta' \in (\beta - \delta, \beta + \delta)$,

$$\lim_{N \rightarrow \infty} F_N(\beta') = \lim_{N \rightarrow \infty} \frac{1}{N} \log Z_N(\beta') = F(\beta'), \text{ a.s.}$$

Theorem

If F is differentiable at β , then

$$\lim_{N \rightarrow \infty} \left\langle \left| \frac{\mathcal{H}_N(\sigma)}{N} - F'(\beta) \right| \right\rangle_{\beta} = 0 \quad \text{a.s.}$$

Equivalence of ensembles

Under differentiability of the limiting free energy:

$$E(\beta) = F'(\beta)$$

- Gibbs measure concentrates at the level set $E(\beta) = F'(\beta)$.
- If F is not differentiable, the Hamiltonian stays inside the interval given by the left and right derivatives at β .
- One tries to understand the geometry of the level set $E(\beta)$ by understanding the Gibbs measure at inverse temperature β .

Thermodynamical approach

- Sample σ and σ' independently from the Gibbs measure G_N .
Goal: understand the limiting distribution of the distance

$$d(\sigma_1, \sigma_2) \text{ under } G_N^2.$$

- Fix $A \subset \mathbb{R}$, look at the constrained coupled free energy:

$$\int_{\Sigma \times \Sigma} \mathbf{1}_{\{d(\sigma_1, \sigma_2) \in A\}} e^{\beta(\mathcal{H}(\sigma_1) + \mathcal{H}(\sigma_2))} d\sigma_1 d\sigma_2.$$

Example 1: O'Connell-Yor semi-discrete polymer model

- Consider the configuration space

$\Sigma_N = \{\sigma \in \mathbb{R}^{N+1} : 0 = \sigma_0 \leq \dots \leq \sigma_{N-1} \leq \sigma_N = N.\}$ Set

$$\mathcal{H}(\sigma) = \sum_{i=1}^N (B_i(\sigma_i) - B_i(\sigma_{i-1})),$$

where $(B_i)_{1 \leq i \leq N}$ are i.i.d. standard Brownian motions.

- Free energy explicit given by:

$$F(\beta) = -\mathcal{L}[-\psi](-\beta^2) - \log(\beta^2), \quad \beta \neq 0,$$

where ψ is the digamma function, and $\mathcal{L}[-\psi]$ is the Legendre transform of $-\psi$:

$$\mathcal{L}[-\psi](x) = \sup_{\theta > 0} [x\theta + \psi(\theta)].$$

Example 1: O'Connell-Yor semi-discrete polymer model

- O'Connell-Yor, Moriarty-O'Connell, Seppalainen-Valko, Borodin-Corwin-Ferrari, Borodin-Corwin, ...
- Explicit derivative in this case:

$$F'(\beta) = 2 \left(\beta \psi_1^{-1}(\beta^2) - \frac{1}{\beta} \right).$$

- Differentiability also expected for other polymer models.

Example 2: Spin glasses

- Spin Glasses are alloys with strange magnetic properties. Ex: CuMn
 - Physically: Spin + Glass
 - Mathematically: Randomness + Frustration
- Spin glasses exhibit many crucial features commonly shared in many real world problems:
 - Traveling salesman problem.
 - Hopfield neural network.

Example 2: The Sherrington-Kirkpatrick (SK) model

- The SK model is defined by the Hamiltonian or energy

$$\mathcal{H}_N(\sigma) = \frac{1}{\sqrt{N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j \quad \left(= \frac{1}{\sqrt{N}} (g\sigma, \sigma) \right)$$

for each $\sigma \in \{-1, +1\}^N$, where g_{ij} are i.i.d. $N(0, 1)$.

- **Frustration:**



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for each $\sigma \in \{-1, +1\}^N$, where g_{ij} are i.i.d. $N(0, 1)$.

- It is a Gaussian process with

$$\mathbb{E} \mathcal{H}_N(\sigma^1) \mathcal{H}_N(\sigma^2) = N(R(\sigma^1, \sigma^2))^2,$$

where

$$R(\sigma^1, \sigma^2) = \frac{1}{N} \sum_{i=1}^N \sigma_i^1 \sigma_i^2.$$

Example 2: The mixed p -spin

- Mixed p -spin model: A Gaussian process with

$$\mathbb{E}\mathcal{H}_N(\sigma^1)\mathcal{H}_N(\sigma^2) = N\xi\left(R(\sigma^1, \sigma^2)\right),$$

where

$$R(\sigma^1, \sigma^2) = \frac{1}{N} \sum_{i=1}^N \sigma_i^1 \sigma_i^2.$$

- SK: $\xi(t) = t^2$.
- Pure p -spin: $\xi(t) = t^p$.

The limit free energy - Parisi Formula

Theorem (M. Talagrand '06, Panchenko '10 - Parisi Formula)

For all $\beta > 0$ the following limit exists a.s.:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log Z_N = F(\beta)$$

- $F(\beta) = F(\beta, \xi)$ is given by a variational principle that for large β is highly untrivial:

$$F(\beta) = \inf_{\rho \in \mathcal{M}[0,1]} \mathcal{P}(\beta, \rho).$$

- Predicted by G. Parisi.
- What is the minimizer? What is the role of the minimizer?

The minimizer of the Parisi Functional

Parisi measure.

We call a measure that minimizes the previous variational problem a *Parisi measure*. We denote it by $\mu_P = \mu_P(\beta)$.

- (Guerra '03) Minimizer exists.
- (A.-Chen '14) Minimizer is unique.

It is expected that:

- Up to symmetry, μ_P is the limiting law of the overlap under $\mathbb{E}G_N^2$.
- The limit law of overlap fully describes the measure G_N as N goes to infinity.

Predictions

More predictions

For any mixed p -spin, any Parisi measure should satisfy the following:

- (1) The origin is contained in the support of the Parisi measure at any temperature.
- (2) One expects FRSB behavior at low temperature.
- (3) Any Parisi measure has a jump discontinuity at the top of the support at any temperature.

Significance of the Parisi measure

Two major predictions:

(1) Full Replica Symmetry Breaking:

SK-known.pdf

(2) Ultrametricity: with probab. ≈ 1 , for i.i.d. $\sigma^1, \sigma^2, \sigma^3 \sim G_N$,

$$\|\sigma^1 - \sigma^2\| \leq \max(\|\sigma^1 - \sigma^3\|, \|\sigma^2 - \sigma^3\|).$$

tree.pdf

Energy landscape

- Can we say anything about the energy landscape of $\mathcal{H}_N$?
- In the bulk (REM Universality): Bovier, Ben Arous-Kuptsov.
- Near the global maximum/minimum: landscape is highly non-trivial.

Spherical model

- **Spherical** mixed p -spin model,

$$\mathcal{H}_N(\sigma) = \sum_{p \in \mathbb{N}} c_p^{1/2} \left(\frac{1}{N^{(p-1)/2}} \sum_{1 \leq i_1, \dots, i_p \leq N} g_{i_1, \dots, i_p} \sigma_{i_1} \cdots \sigma_{i_p} \right)$$

for $\sigma \in S^{N-1} := \{\sigma \in \mathbb{R}^N : \sum_{i=1}^N \sigma_i^2 = N\}$.

- Note

$$\mathbb{E} \mathcal{H}_N(\sigma^1) \mathcal{H}_N(\sigma^2) = N \xi(R(\sigma^1, \sigma^2)),$$

where

$$\xi(x) = \sum_{p \in \mathbb{N}} c_p s^p \text{ for } \sum_{p \in \mathbb{N}} c_p = 1.$$

Example: Spherical SK model

- Write

$$\mathcal{H}_N(\sigma) = \frac{1}{\sqrt{N}} \sum_{i,j=1}^N g_{ij} \sigma_i \sigma_j = \frac{1}{\sqrt{N}} (g\sigma, \sigma) = \frac{1}{\sqrt{2N}} (W\sigma, \sigma),$$

where W is a GOE matrix, i.e., $W_{ij} = W_{ji}$, $\mathbb{E}W_{ij} = 0$, $\mathbb{E}W_{ij}^2 = 1$.

- Then

$$\max_{\sigma \in S^{N-1}} \mathcal{H}_N(\sigma) = \frac{\text{Largest eigenvalue of } W}{\sqrt{2N}}.$$

- The critical points of H_N are $\pm$ eigenvectors of W .
- Easy to describe the landscape: 2 local minima, 2 saddle points of any index.

Energy landscape: near the minimum

energylandscape.jpg

Energy landscape: multiple peaks

$$\text{Overlap } R(\sigma, \sigma') = \frac{1}{N} \sum_{i=1}^N \sigma_i \sigma'_i.$$

Theorem (Multiple peaks, A.-Chen '17)

For any $\varepsilon > 0$, there exists a constant $K > 0$ s.t. for any $N \geq 1$, with probability at least $1 - Ke^{-N/K}$, $\exists S_N \subset \Sigma_N$ such that

- (i) $|S_N| \geq e^{N/K}$,
- (ii) $\forall \sigma \in S_N, \left| \frac{\mathcal{H}_N(\sigma)}{N} - \min_{\sigma' \in \Sigma_N} \frac{\mathcal{H}_N(\sigma')}{N} \right| < \varepsilon$.
- (iv) $\forall \sigma, \sigma' \in S_N$ with $\sigma \neq \sigma'$, $|R(\sigma, \sigma')| < \varepsilon$.

- (1) Chatterjee '09: $|S_N| \geq (\log N)^c$
- (2) Ding-Eldan-Zhai '14: $|S_N| \geq N^c$.
- Chen-Handschy-Lerman '18: $\Sigma_N = \{\pm 1\}^N$.
- (1) and (2): larger class of Gaussian Processes

Sphere: Crisanti-Sommers formula

Theorem (Chen-Sen '15, Jagannath-Tobasco '16)

$$\begin{aligned} GSE &:= \lim_{N \rightarrow \infty} \min_{\sigma \in S_N} \left(\frac{\mathcal{H}_N(\sigma)}{N} + h \frac{\sum_{i=1}^N \sigma_i}{N} \right) \\ &= \inf_{\nu} \frac{1}{2} \left(\int_0^1 (\xi'(s) + h^2) \nu(ds) + \int_0^1 \frac{dq}{\nu((q, 1])} \right), \end{aligned}$$

where the infimum is over all $\nu(ds) = \gamma(s)1_{[0,1)}(s)ds + \Delta\delta_1(ds)$.

- Minimizer exists and is unique. We call ν_P the Parisi measure.
- Denote

$$\nu_P(ds) = 1_{[0,1)}(s)\gamma_P(s)ds + \Delta_P\delta_1(s).$$

- What does γ_P look like?
 - **Replica Symmetry (RS):** $\gamma_P \equiv 0$
 - **One Replica Symmetry Breaking (1RSB):** $\gamma_P \equiv C > 0$
 - **Full Replica Symmetry Breaking (FRSB):** $\gamma_P(ds) = f(s)ds$

1 RSB scenario

Theorem

Assume $h = 0$ and $\xi(s) = s^p$ for even $p \geq 3$. For any $\varepsilon > 0$, there exist $\eta, K > 0$ such that for all $N \geq 1$,

$$\mathbb{P}(\exists \sigma^1, \sigma^2 \in \mathcal{L}_N(\eta) \text{ s.t. } |R(\sigma^1, \sigma^2)| \in [\varepsilon, 1 - \varepsilon]) \leq Ke^{-\frac{N}{K}}. \quad (1)$$

1 RSB phase

- No path of low energy connecting different “components”.
Existence of diverging barriers.
- Number of connected components is exponentially large in N :
same applies to the Euler characteristic of the level set.
- A.-Ben Arous-Cerny '13 + Subag '15 $\Rightarrow$ similar results for pure p -spin by counting the number of critical points.
- Theorem above holds for a large collection of ξ 's, e.g.,

$$\xi(s) = cs^p + (1 - c)s^{p+2} \text{ for even } p \geq 4.$$

$$\xi(s) = \frac{e^{s^2} - 1 - s^2}{e - 2}.$$

Energy landscape: Full Replica Symmetry Breaking

Theorem

If $\xi'(1) + h^2 < \xi''(1)$ and $\xi''(s)^{-1/2}$ is concave on $(0, 1]$, then

$$\gamma_0(s) = \begin{cases} 0, & \text{if } q \in [0, q_0), \\ \frac{\xi'''(s)}{2\xi''(s)^{3/2}}, & \text{if } q \in [q_0, 1) \end{cases}$$

and

$$GSE = q_0 \xi''(q_0)^{1/2} + \int_{q_0}^1 \xi''(q)^{1/2} dq,$$

where $q_0 \in [0, 1]$ is the unique solution to $\xi'(q_0) + h^2 = q_0 \xi''(q_0)$.

Energy landscape: Full Replica Symmetry Breaking

Theorem (A. - Chen '17)

Let $h \neq 0$. Assume $\xi'(1) + h^2 < \xi''(1)$ and $\frac{1}{\sqrt{\xi''}}$ is concave on $(0, 1]$.

- ① Let $u \in [q_0, 1]$. For any $\varepsilon, \eta > 0$, there exists $K > 0$ such that

$$\mathbb{P}(\exists \sigma^1, \sigma^2 \in \mathcal{L}_N(\eta) \text{ s.t. } |R(\sigma^1, \sigma^2) - u| < \varepsilon) \geq 1 - Ke^{-\frac{N}{K}}.$$

- ② For any $\varepsilon > 0$, there exist $\eta, K > 0$ such that

$$\mathbb{P}(\exists \sigma^1, \sigma^2 \in \mathcal{L}_N(\eta) \text{ s.t. } R(\sigma^1, \sigma^2) \in [-1, q_0 - \varepsilon]) \leq Ke^{-\frac{N}{K}}.$$

Main ingredients of the proof

- Guerra-Talagrand's bound: control of the constrained partition function.
- Stochastic optimization: handle the behavior of the solution of Parisi's formula.
- Connect the support of the Parisi measure to the possible distances between points in the level set.

Open questions

- SK model:
 - Quantitative dependence of the energy landscape on N .
 - Further structure of the energy landscape.
 - Fluctuation.
- Spherical model:
 - A.- Zeng '18: Examples of 2-RSB models.
 - Determine the phase diagram for the energy landscape.
- Bipartite SK model: Let $N_1 = cN$ and $N_2 = (1 - c)N$.

$$H_N(\sigma) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} g_{ij} \tau_i \rho_j$$

for $\sigma = (\tau, \rho) \in \{-1, +1\}^{N_1} \times \{-1, +1\}^{N_2}$. Note

$$\mathbb{E} H_N(\sigma) H_N(\sigma') = c(1 - c) N R(\tau, \tau') R(\rho, \rho').$$

Questions:

- Free energy?
- Maximum energy?
- Energy landscape?

Thank you for your attention.